

DSA + LLD Combined Prep Roadmap

Atin Sharma · SDE-1 Targets · 8-Week Parallel Plan · TUF+ · Educative · LeetCode

■ Must — do every problem ■ High — do if time ■ Good — skip for SDE-1 ■ Strike — explicitly skipped

Part 1 — DSA Roadmap (Post-Trees, Next 3–4 Weeks)

Resource: TUF+ A2Z Sheet (primary) + LeetCode for submission

You have completed Arrays/HashMap, and most of Trees. Everything below is pending. Problems are ordered by pattern sequence — complete ■ Must problems in every pattern before touching ■ High. ■ Good problems are explicitly skipped at SDE-1 bar.

Concept Sequence + SDE-1 Importance

Pattern	Must / High / Skip	Core Concept to Lock In	Platform
Two Pointers	5 Must + 1 High	Shrink from both ends; sorted array + pair sum	TUF+ A2Z → LC
Sliding Window	4 Must + 1 High	Expand/shrink window; freq map pattern	TUF+ A2Z → LC
Binary Search	3 Must + 2 High	Which half to eliminate; binary on answer	TUF+ A2Z → LC
Linked List	6 Must + 1 High	Fast/slow pointers; dummy head pattern	TUF+ A2Z → LC
Stack	5 Must	Monotonic stack for NGE; aux stack for MinStack	TUF+ A2Z → LC
Trees (finish)	3 Must + 2 High	BST property (inorder = sorted); LCA walk	TUF+ A2Z → LC
Graphs	7 Must + 2 High	BFS grid; DFS connected; Topo sort (Kahn's)	TUF+ A2Z → LC
Heap / PQ	4 Must + 2 High	K-element window; two-heap median	TUF+ A2Z → LC
Backtracking	6 Must + 2 High	Decision tree; pruning via constraints	TUF+ A2Z → LC
Greedy	6 Must + 1 High	Local optimal = global; interval sort by end	TUF+ A2Z → LC
DP 1D	9 Must + 1 High	Write recurrence BEFORE code; tabulation > memo	TUF+ A2Z → LC
DP 2D	3 Must + 1 High	State transition table; LCS/Edit Distance template	TUF+ A2Z → LC

Explicit Skip List — Do Not Touch at SDE-1 Bar

Problem	Difficulty	Priority	Reason
4Sum	Medium	■ High	Low frequency — skip
First Missing Positive	Hard	■ Good	Skip for SDE-1
Median of Two Sorted Arrays	Hard	■ Good	Skip for SDE-1
Allocate Books / Split Array	Hard	■ High	Skip — advanced BS on answer
IPO	Hard	■ Good	Skip
N-Queens	Hard	■ High	Skip — very low SDE-1 frequency
Sudoku Solver	Hard	■ Good	Skip
Candy	Hard	■ High	Low frequency
Interleaving String	Hard	■ Good	Skip — DP 2D niche
Burst Balloons	Hard	■ Good	Skip — DP interval niche
Serialize/Deserialize Binary Tree	Hard	■ Good	Skip
Network Delay Time (Dijkstra)	Medium	■ High	Skip — SDE-2 graph algo
Word Ladder	Hard	■ High	Skip — niche BFS

Per-Pattern Problem List — Must Problems

Two Pointers	Two Pointers	5 Must problems
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LC #	Problem	Difficulty	Note
125	Valid Palindrome	Easy	Classic entry
15	3Sum	Medium	Sort + 2-ptr
11	Container With Most Water	Medium	Greedy shrink
42	Trapping Rain Water	Hard	Left/right max
167	Two Sum II (Input Sorted)	Medium	Binary + 2ptr

Sliding Window	Sliding Window	4 Must problems	
LC #	Problem	Difficulty	Note
3	Longest Substring No Repeat	Medium	In Progress—finish
567	Permutation in String	Medium	In Progress—finish
76	Minimum Window Substring	Hard	Variable window
424	Longest Repeating Char Replacement	Medium	Shrink on invalid

Binary Search	Binary Search	3 Must problems	
LC #	Problem	Difficulty	Note
704	Binary Search Classic	Easy	Template anchor
33	Search in Rotated Sorted Array	Medium	Which half?
153	Find Min in Rotated Sorted Array	Medium	Variant of above

Linked List	Linked List	6 Must problems	
LC #	Problem	Difficulty	Note
21	Merge Two Sorted Lists	Easy	Pointer weave
141	Linked List Cycle	Easy	Floyd fast/slow
876	Middle of Linked List	Easy	Fast/slow
19	Remove Nth Node From End	Medium	Two-pass / gap
142	Linked List Cycle II	Medium	Entry of cycle
143	Reorder List	Medium	Mid + reverse + merge

Stack	Stack	5 Must problems	
LC #	Problem	Difficulty	Note
20	Valid Parentheses	Easy	Bracket match
155	Min Stack	Medium	Aux stack trick
496	Next Greater Element I	Easy	Mono stack intro
739	Daily Temperatures	Medium	Mono stack
84	Largest Rectangle in Histogram	Hard	Mono stack hard

Trees	Trees (remaining)	5 Must problems	
LC #	Problem	Difficulty	Note
230	Kth Smallest in BST	Medium	Inorder = sorted
235	LCA of BST	Medium	BST property walk
98	Validate BST	Medium	Min/max bounds
124	Binary Tree Max Path Sum	Hard	Post-order gain

LC #	Problem	Difficulty	Note
105	Construct from Pre+Inorder	Medium	Divide & recurse

Graphs		7 Must problems	
LC #	Problem	Difficulty	Note
200	Number of Islands	Medium	DFS/BFS grid
133	Clone Graph	Medium	HashMap + DFS
695	Max Area of Island	Medium	DFS + count
994	Rotting Oranges	Medium	Multi-source BFS
207	Course Schedule	Medium	Topo sort / cycle
210	Course Schedule II	Medium	Topo order
323	Number of Connected Components	Medium	Union-Find or BFS

Heap / Priority Queue		4 Must problems	
LC #	Problem	Difficulty	Note
215	Kth Largest Element in Array	Medium	Min-heap size K
973	K Closest Points to Origin	Medium	Max-heap size K
621	Task Scheduler	Medium	Greedy + heap
295	Find Median from Data Stream	Hard	Two heaps

Backtracking		6 Must problems	
LC #	Problem	Difficulty	Note
78	Subsets	Medium	Include/exclude
46	Permutations	Medium	Swap approach
39	Combination Sum	Medium	Repeat allowed
40	Combination Sum II	Medium	No repeat
79	Word Search	Medium	DFS on grid
22	Generate Parentheses	Medium	Open/close count

Greedy		6 Must problems	
LC #	Problem	Difficulty	Note
55	Jump Game	Medium	Max reach
45	Jump Game II	Medium	Min jumps
134	Gas Station	Medium	Net tank
252	Meeting Rooms I	Easy	Sort overlap
435	Non-overlapping Intervals	Medium	Sort + greedy
56	Merge Intervals	Medium	Sort + merge

DP 1D		9 Must problems	
LC #	Problem	Difficulty	Note
70	Climbing Stairs	Easy	Fib recurrence
198	House Robber	Medium	Max no-adj
213	House Robber II	Medium	Circle variant

LC #	Problem	Difficulty	Note
322	Coin Change	Medium	Unbounded knapsack
300	LIS	Medium	dp[i] + patience
5	Longest Palindromic Substring	Medium	Expand around center
91	Decode Ways	Medium	1 or 2 digit
139	Word Break	Medium	dp + dict lookup
152	Max Product Subarray	Medium	Track min/max

DP 2D DP 2D 3 Must problems			
LC #	Problem	Difficulty	Note
62	Unique Paths	Medium	Grid dp
1143	Longest Common Subsequence	Medium	Classic 2D dp
72	Edit Distance	Hard	Insert/del/replace

Part 2 — LLD Roadmap (Strict SDE-1 Bar, No HLD)

Resource: TUF+ LLD Course (primary) + Educative Grokking LLD (case studies)

Universal Drill Protocol — Apply to Every Problem

Step	Action	Time	Rule
1	Read problem requirements on TUF+	5 min	Read once, no notes
2	Close TUF+. Draw class diagram on Excalidraw from memory	15 min	Name every class and arrow type
3	Write full TypeScript classes — attributes, methods, interfaces	30 min	No peeking at diagram
4	Reopen TUF+. Compare. Write 3-bullet gap list	10 min	What did you miss?
5	Re-do the entire problem from scratch 2 days later	45 min	Cold recall — no notes

Phase 1 Foundations + Creational Patterns

Weeks 1–2 · 45–60 min/day

Intro to LLD + Software Design Principles

Platform: TUF+ · TUF+ LLD course, Lesson 1–2

- 1 Watch video → pause at every example → redraw the class structure on paper
- 2 Write 2-line summary of why each principle exists before moving on

SOLID Principles (SRP, OCP, LSP, ISP, DIP)

Platform: TUF+ · TUF+ LLD, Lessons 3–7

- 1 Watch lesson → close tab → write a TypeScript violation example from memory
- 2 Write the corrected version and explain what changed in a comment
- 3 10 min per principle — do not over-theorise

UML + Class Diagrams

Platform: TUF+ · TUF+ LLD, Lessons 8–9

- 1 Watch class diagram lesson → recreate the example diagram from scratch on Excalidraw
- 2 Add inheritance, composition, and association arrows without looking at the video

Creational Patterns: Singleton, Factory, Builder, Abstract Factory

Platform: TUF+ · TUF+ LLD, Lessons 10–14

- 1 Watch video → write pattern from scratch in TypeScript, not Java
- 2 Write one 2-line real-world analogy (e.g. Builder = Burger order builder)
- 3 For Singleton: implement thread-safe variant. For Factory: use a string-keyed registry
- 4 Skip Prototype for now — low SDE-1 frequency

Phase 2 Structural + Behavioural Patterns

Weeks 3–4 · 45–60 min/day

Structural: Adapter, Decorator, Facade, Proxy

Platform: TUF+ · TUF+ LLD, Lessons 16–20

- 1 Watch → write from scratch in TS → test with a short main() block
- 2 For Decorator: chain 2 decorators (e.g. logger + caching around a service)
- 3 Skip Composite, Bridge, Flyweight — SDE-2 territory

Behavioural: Observer, Strategy, Command, State, Iterator

Platform: TUF+ · TUF+ LLD, Lessons 23–27

- 1 Observer: implement EventEmitter from scratch — no library
- 2 Strategy: model a payment processor that swaps UPI/Card/Wallet at runtime
- 3 Command: implement Undo/Redo with a command stack
- 4 State: model a TrafficLight with 3 states — used directly in Traffic Signal problem
- 5 Skip Template Method, Chain of Responsibility, Visitor, Mediator, Memento

Phase 3

Advanced Topics (Selective Reads)

Week 5 · 30–45 min/day

Dependency Injection + Exception Handling + Best Practices + API Design

Platform: TUF+ · TUF+ LLD, Advanced section

- 1 Read (not watch) — these are 1-day reference reads, not deep practice topics
- 2 Note: DI connects directly to how you structured VyaparGPT's service layer
- 3 Exception Handling: write a typed Result monad — you already use this pattern
- 4 API Design: focus on REST contract design, idempotency keys, versioning

SKIP: Multithreading + Concurrency (entire module)

Platform: TUF+ · Skip — revisit after SDE-1 offers

- 1 Full module is SDE-2+ territory. At SDE-1 you may get 1 conceptual question max
- 2 If asked: know the definition of deadlock, race condition, and mutex — 5 minutes of reading

Phase 4

Interview Problems — Design + Code

Weeks 6–8 · 1 problem per 2–3 days

Tier 1: Parking Lot, Vending Machine, Logging Framework

Platform: TUF+ · TUF+ LLD Part-1

- 1 STEP 1 — Read requirements (5 min). Close TUF+.
- 2 STEP 2 — Draw class diagram from scratch on Excalidraw (15 min).
- 3 STEP 3 — Write full TypeScript classes with methods and attributes (30 min).
- 4 STEP 4 — Reopen TUF+. Compare with solution. Write a 3-bullet gap list.
- 5 STEP 5 — Re-do the same problem from scratch after 2 days without notes.
- 6 Parking Lot: uses Strategy (pricing), Factory (slot allocation), Observer (availability alerts)
- 7 Vending Machine: uses State pattern — 4 states: Idle, HasMoney, Dispensing, OutOfStock

Tier 1: ATM, Task Manager, PubSub System

Platform: TUF+ · TUF+ LLD Part-2

- 1 ATM uses State pattern — explicitly name it during the design step
- 2 PubSub maps to your BullMQ + Upstash Redis work — use that mental model directly
- 3 Task Manager: uses Observer (status change) + Command (undo task)
- 4 Same 5-step drill protocol as Tier 1 above

Tier 2: Elevator, Ride Booking, Hotel Management, Traffic Signal

Platform: TUF+ · TUF+ LLD Part-3 + Educative Grokking LLD Chapters 7,8,11,16

- 1 Elevator: uses State (Idle/Moving/Door-open) + Strategy (scheduling: SCAN vs LOOK)
- 2 Ride Booking: uses Observer (driver notified) + Strategy (matching) + Factory (ride types)
- 3 Traffic Signal: uses State pattern directly — you built the State concept in Phase 2
- 4 Hotel Management: uses Factory (room types) + Observer (booking alerts)
- 5 Also do these from Educative for extra diagram depth: Elevator, ATM, Parking Lot, Vending Machine

Part 3 — Combined Daily Morning Schedule (8 Weeks)

DSA: 6:00–8:30 AM with Srishti (45-min timed blocks) · LLD: Evening 9:00–10:00 PM

■ DSA morning block is locked — 6:00–8:30 AM, non-negotiable, with timed 45-min blocks. LLD evening block is flexible — 45–60 min on weekdays, 90 min on weekends for problem drills. If DP or Graph week is heavy, cut LLD to 30 min or skip 1 day — protect DSA time.

Week / Phase	Time Block	Activity	Platform	Drill Detail
Week 1 (Trees finish + Two Ptr start)	6:00–7:30 AM	DSA — Trees finish	TUF+ A2Z	Complete remaining Tree problems: LC230, 235, 98, 105. 45-min timed block with Srishti.
	7:30–8:30 AM	DSA — Two Pointers	TUF+ A2Z	LC125 Valid Palindrome + LC15 3Sum. 1 problem per session minimum.
	9:00–10:00 PM	LLD — Foundations	TUF+ LLD	Watch SOLID lessons (SRP, OCP today). Write 1 violation + 1 fix per principle in TS.
Week 2 (Sliding Win + Binary Search)	6:00–7:30 AM	DSA — Sliding Window	TUF+ A2Z	Finish LC3 + LC567 (In Progress). Then LC76 Minimum Window Substring.
	7:30–8:30 AM	DSA — Binary Search	TUF+ A2Z	LC704 → LC33 → LC153. Focus: which-half elimination logic.
	9:00–10:00 PM	LLD — SOLID finish	TUF+ LLD	Complete LSP, ISP, DIP. Draw UML class diagram for a sample system on Excalidraw.
Week 3 (Linked List + Stack)	6:00–7:30 AM	DSA — Linked List	TUF+ A2Z	LC21, LC141, LC876, LC142. Floyd cycle algo — implement both detection and entry.
	7:30–8:30 AM	DSA — Stack	TUF+ A2Z	LC20 Valid Parentheses → LC155 Min Stack → LC496 Next Greater Element.
	9:00–10:00 PM	LLD — Creational Patterns	TUF+ LLD	Singleton + Factory today. Builder + Abstract Factory tomorrow. 1 pattern per night.
Week 4 (Graphs + Heap)	6:00–7:30 AM	DSA — Graphs (BFS/DFS)	TUF+ A2Z	LC200 → LC695 → LC994. Core template: visited set + BFS queue or DFS stack.
	7:30–8:30 AM	DSA — Graphs (Topo Sort) + Heap	TUF+ A2Z	LC207 + LC210 Topo sort. Then LC215 Kth Largest (heap warm-up).
	9:00–10:00 PM	LLD — Structural Patterns	TUF+ LLD	Adapter + Decorator. Implement Decorator: chain Logger + Cache around a service class.
Week 5 (Backtracking + Greedy)	6:00–7:30 AM	DSA — Heap finish + Backtracking	TUF+ A2Z	LC973 + LC621 + LC295. Then start Backtracking: LC78 Subsets using include/exclude tree.
	7:30–8:30 AM	DSA — Backtracking + Greedy	TUF+ A2Z	LC46 Permutations + LC39 Combination Sum. Greedy: LC55 Jump Game + LC56 Merge Int.
	9:00–10:00 PM	LLD — Behavioural Patterns	TUF+ LLD	Observer + Strategy. Build EventEmitter from scratch. Swap strategy demo.
Week 6 (DP 1D + LLD Problems)	6:00–7:30 AM	DSA — DP 1D	TUF+ A2Z	LC70 → LC198 → LC213 → LC322 Coin Change. Key: write recurrence BEFORE code.
	7:30–8:30 AM	DSA — DP 1D cont.	TUF+ A2Z	LC300 LIS + LC5 Palindromic Substring + LC91 Decode Ways.
	9:00–10:00 PM	LLD — Parking Lot + Vending Machine	TUF+ LLD	Full 5-step drill: requirements → diagram → code → compare → re-do in 2 days.
Week 7 (DP 2D + LLD Problems)	6:00–7:30 AM	DSA — DP 2D	TUF+ A2Z	LC62 Unique Paths → LC1143 LCS → LC72 Edit Distance. Write state transition table first.
	7:30–8:30 AM	DSA — Revision / weak pattern	TUF+ A2Z	Identify your lowest-scored pattern from Discord sessions. Redo 2 problems from that pattern.
	9:00–10:00 PM	LLD — ATM + PubSub	TUF+ LLD	ATM: name the State pattern explicitly. PubSub: draw the subscriber registry structure.
Week 8 (Mock Week)	6:00–7:30 AM	DSA Mock — 2 problems	leetcode	1 Medium from a weak pattern + 1 Medium from Graphs or DP. No hints. Score yourself.
	7:30–8:30 AM	DSA — Remaining High priority	TUF+ A2Z	Clear any remaining ■ High problems: LC146 LRU Cache, LC417 Pacific Atlantic, LC57 Insert
	9:00–10:00 PM	LLD — Elevator + Ride Booking + Elevator	TUF+ LLD	State machine: design State machine diagram first. Ride Booking: anchor to VyaparGPT routing.

Platform Quick Reference

Task	Primary Resource	Where to Find It	Secondary
DSA concept videos	TUF+	tuf+.org → A2Z Sheet → respective pattern section	YouTube TUF channel (free)
DSA problem submissions	leetcode	leetcode.com → problem # directly	—
LLD concept videos	TUF+	tuf+.org → LLD Course → numbered lesson	—
LLD case study diagrams	Grokking	Grokking LLD Interview Using OOD → chapter by system	TUF+ for code
LLD mock interviews	Educative	Grokking LLD → Mock Interview at end of each chapter	Discord partner
OOP fundamentals	Educative	Grokking LLD Ch. 2–5 (read, not watch)	TUF+ video if unclear
Design patterns reference	refrence	LLD Course Lessons 10–31	refactoring.guru (free)

Hard Rules — Non-Negotiable

1. DSA morning block (6–8:30 AM) is never cut — it is the primary interview signal.
2. DP week (Week 6–7): if both sessions exceed capacity, drop LLD to 30 min that night — not DSA.
3. Graph week (Week 4): LLD evening is structural pattern week — 45 min is sufficient, no extension needed.
4. Do not touch ■ Good problems until you have solved every single ■ Must problem across all patterns.
5. Re-do ToDoAgain problems (Group Anagrams LC49, Top K Elements LC347, Longest Consecutive LC128) in Week 1 alongside Trees.
6. LLD problems: never skip the Step 5 re-do. The cold recall is the practice. Reading is not practice.
7. For every LLD pattern you encounter, name it out loud during the design step — interviewers reward vocabulary.